Question ID: 49800634

Question

x	y
-18	-48
7	52

The table shows two values of x and their corresponding values of y. In the *xy*-plane, the graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{7},\ a\right)$. What is the value of a?

Answer

A. $-\frac{4}{11}$

B. $-\frac{4}{77}$

C. $\frac{4}{7}$

D. $\frac{172}{7}$

Question ID: a1fd2304

Question

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Question ID: a7a14e87

Question

In the xy-plane, line k is defined by $x+y=0$. Line j is perpendicular to line k, and the y-intercept of line j is $(0,3)$. Which of the following is an equation of line j ?

Answer

A. $x+y=3$

B. $x+y=-3$

C. $x-y=3$

D. $x-y=-3$

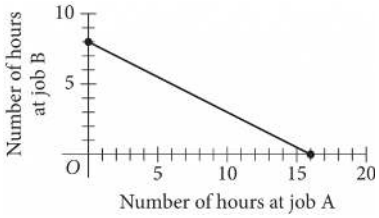
Question ID: cb58833c

Question

The line with the equation $\frac{4}{5}x + \frac{1}{3}y = 1$ is graphed in the xy-plane. What is the x-coordinate of the x-intercept of the line?

Question ID: c4ea43ef

Question



To earn money for college, Avery works two part-time jobs: A and B. She earns $10 per hour working at job A and $20 per hour working at job B. In one week, Avery earned a total of s dollars for working at the two part-time jobs. The graph above represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn s dollars. What is the value of s ?

Answer

A. 128

B. 160

C. 200

D. 320

Question ID: 2d54c272

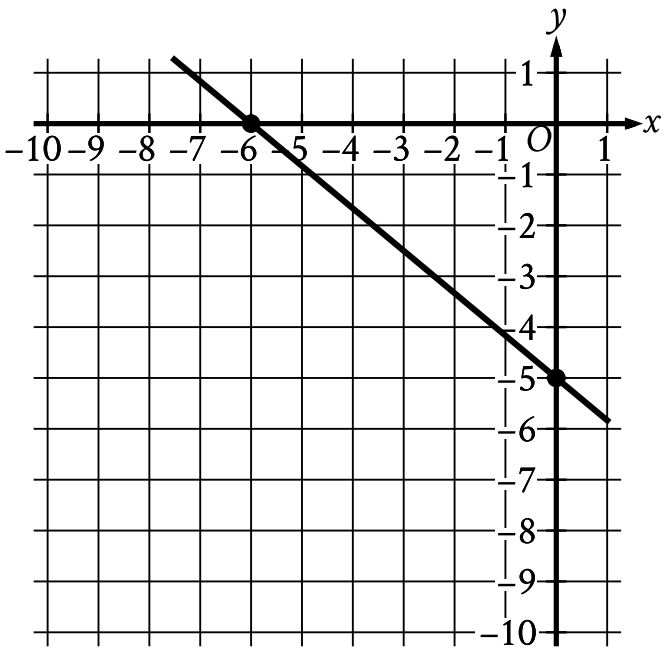
Question

$$5G + 45R = 380$$

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won G green tokens and R red tokens worth a total of 380 points. The given equation represents this situation. How many more points is a red token worth than a green token?

Question ID: 6d8ad460

Question



Line k is shown in the *xy*-plane. Line j (not shown) is perpendicular to line k. What is the slope of line j?

Question ID: 0366d965

Question

x	y
3	7
k	11
12	n

The table above shows the coordinates of three points on a line in the xy-plane, where k and n are constants. If the slope of the line is 2, what is the value of $k+n$?

Question ID: cc3e9528

Question

The graph of $9x - 10y = 19$ is translated down 4 units in the *xy*-plane. What is the *x*-coordinate of the *x*-intercept of the resulting graph?

Question ID: 98d3393a

Question

Line ℓ in the xy-plane is perpendicular to the line with equation $x = 2$. What is the slope of line ℓ?

Answer

A. 0

B. $-\frac{1}{2}$

C. -2

D. The slope of line ℓ is undefined.

Question ID: 94b48cbf

Question

The graph of $7x + 2y = -31$ in the xy-plane has an x-intercept at $(a, 0)$ and a y-intercept at $(0, b)$, where a and b are constants. What is the value of $\frac{b}{a}$?

Answer

A. $-\frac{7}{2}$

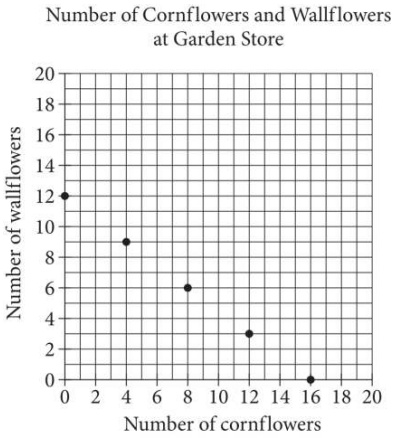
B. $-\frac{2}{7}$

C. $\frac{2}{7}$

D. $\frac{7}{2}$

Question ID: c362c210

Question



The points plotted in the coordinate plane above represent the possible numbers of wallflowers and cornflowers that someone can buy at the Garden Store in order to spend exactly $24.00 total on the two types of flowers. The price of each wallflower is the same and the price of each cornflower is the same. What is the price, in dollars, of 1 cornflower?

Question ID: 0b46bad5

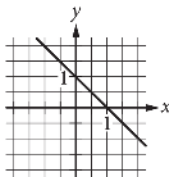
Question

$ax + by = b$

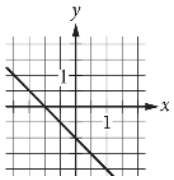
In the equation above, a and b are constants and $0 < a < b$. Which of the following could represent the graph of the equation in the xy-plane?

Answer

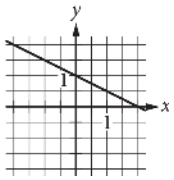
A.



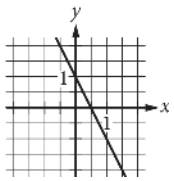
B.



C.

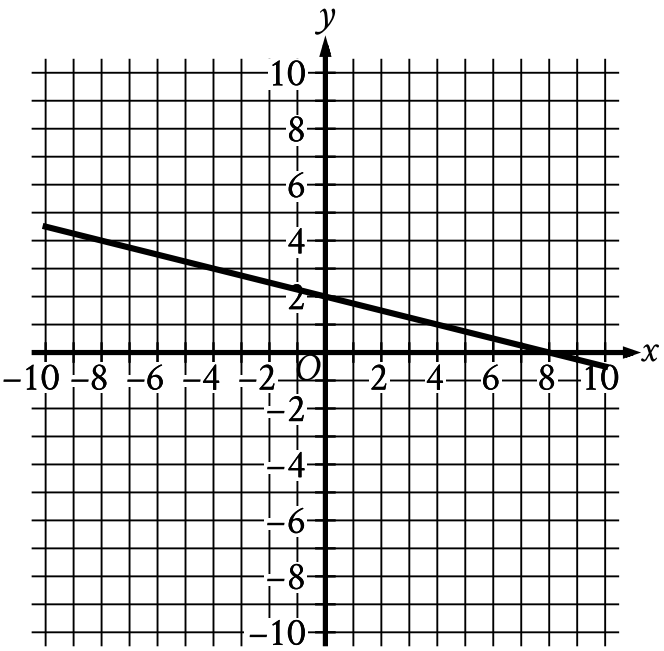


D.



Question ID: 05bb1af9

Question



The graph of $y = f(x) + 14$ is shown. Which equation defines function f?

Answer

A. $f(x) = -\frac{1}{4}x - 12$

B. $f(x) = -\frac{1}{4}x + 16$

C. $f(x) = -\frac{1}{4}x + 2$

D. $f(x) = -\frac{1}{4}x - 14$

Question ID: b9835972

Question

In the *xy*-plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 2$. If line ℓ also passes through the point $(3, d)$, what is the value of d?

Question ID: 9aaf7786

Question

In the *xy*-plane, line p has a slope of $-\frac{5}{3}$ and an *x*-intercept of $(-6, 0)$. What is the *y*-coordinate of the *y*-intercept of line p?

Question ID: fdee0fbf

Question

In the xy-plane, line k intersects the y-axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k, what is the value of w ?

Question ID: db422e7f

Question

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the *xy*-plane. What is the slope of line r?

Question ID: 9bbce683

Question

x	y
18	130
23	160
26	178

For line h, the table shows three values of x and their corresponding values of y. Line k is the result of translating line h down 5 units in the *xy*-plane. What is the *x*-intercept of line k?

Answer

A. $\left(-\frac{26}{3}, 0\right)$

B. $\left(-\frac{9}{2}, 0\right)$

C. $\left(-\frac{11}{3}, 0\right)$

D. $\left(-\frac{17}{6}, 0\right)$

Question ID: 3cdbf026

Question

The graph of the equation $ax + ky = 6$ is a line in the xy-plane, where a and k are constants. If the line contains the points $(-2, -6)$ and $(0, -3)$, what is the value of k ?

Answer

A. -2

B. -1

C. 2

D. 3

Question ID: 3008cfc3

Question

x	y
k	13
$k+7$	-15

The table gives the coordinates of two points on a line in the *xy*-plane. The *y*-intercept of the line is $(k-5, b)$, where k and b are constants. What is the value of b?